AI AGENTS, MCP & APIS FOR PMS AND BAS · LESSON 6 OF 11

Voice Agents and Voice Cloning: Tools, Architecture, Build Steps and Responsible Use

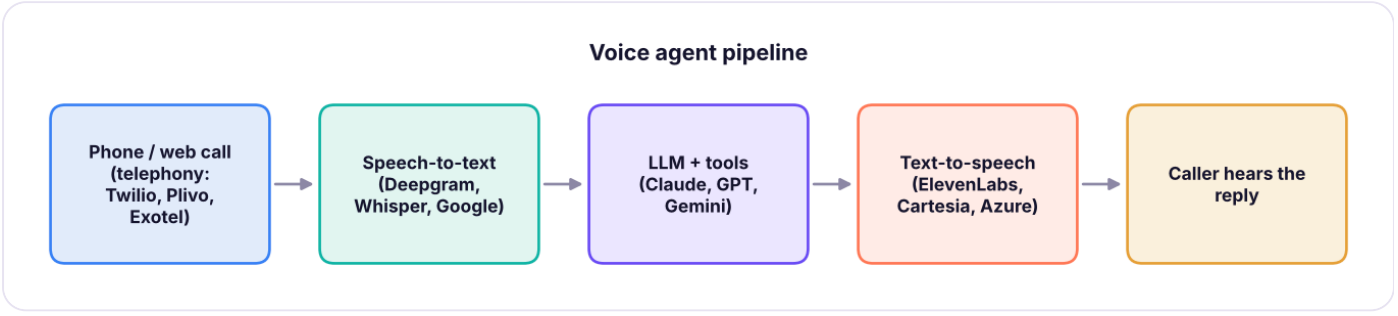
Learn how AI voice agents work (speech-to-text, LLM, text-to-speech, turn-taking, telephony), compare the main platforms (ElevenLabs, Vapi, Retell AI, Bland AI, Synthflow, OpenAI Realtime, Twilio), build a no-code appointment-booking voice agent connected to your calendar and CRM, clone a voice with consent, and design for latency, Indian languages, compliance and quality.

✦ 2026 | Intermediate | ⏱ 120 min

By the end of this lesson you can

- Explain the voice-agent pipeline and why latency matters
- Compare voice-agent and voice-cloning tools and choose one for a use case
- Build a no-code voice agent with tools, a knowledge base and call transfer
- Clone a voice responsibly with consent and disclosure
- Measure and improve a voice agent in production

How a voice agent works



A voice agent pipeline. Speech-to-speech models merge some of these steps.

Conversations feel natural only when the reply starts in well under a second and the agent handles **turn-taking** (knowing when the caller has finished), **barge-in** (stopping when interrupted) and background noise.

Python

```
budget = {"telephony + network": 120, "speech-to-text (streaming)": 200, "LLM first token":
350, "text-to-speech first audio": 150}
total = sum(budget.values())
for k, v in budget.items(): print(f"{k:28s}{v:5d} ms")
print(f"{'time to first audio':28s}{total:5d} ms →", "feels natural" if total ⩽ 1000 else
"feels slow")
```

OUTPUT

```
telephony + network           120 ms
speech-to-text (streaming)    200 ms
LLM first token               350 ms
text-to-speech first audio    150 ms
time to first audio           820 ms → feels natural
```

Ways to cut latency: streaming at every step, a faster model for the conversation (a bigger one only for hard cases), short replies, prompt caching, and servers close to your callers (India region where available).

Platforms

Tool	What it is best for
ElevenLabs	Very natural text-to-speech, voice cloning, dubbing; **ElevenLabs Agents** platform for voice agents with tools and knowledge
Vapi	Developer-friendly voice-agent platform: bring your own STT/LLM/TTS, tools, phone numbers, squads of agents
Retell AI	Production phone agents with a visual flow builder, call analytics and integrations
Bland AI	Enterprise outbound/inbound phone agents with conversational pathways
Synthflow	No-code voice agents for businesses, CRM and calendar integrations
OpenAI Realtime API, **Gemini Live**	Speech-to-speech models for low-latency custom builds
Twilio / Plivo / Exotel	Telephony: numbers, call routing, SIP (Exotel and Plivo are common in India)
Deepgram, AssemblyAI	Fast streaming speech-to-text
Cartesia, PlayHT, Azure AI Speech, Google TTS	Text-to-speech options, including Indian-language voices

Choose by: languages and accents you need (Hindi, Hinglish, Tamil...), latency, telephony in your country, integrations, analytics, compliance, and cost per minute.

Build a no-code appointment-booking voice agent

STEP BY STEP **Clinic receptionist agent (Retell, Vapi, Synthflow or ElevenLabs Agents)**

1. **Prompt**: role ("receptionist for Sunrise Dental, Gurgaon"), goal (book, reschedule, answer FAQs), rules (never give medical advice; transfer emergencies), tone (warm, short sentences), languages (English and Hindi).
2. **Knowledge base**: upload timings, address, services and prices, insurance policy.
3. **Tools**: `check_availability` and `book_appointment` connected to Google Calendar or Cal.com (many platforms have native integrations; otherwise call an n8n/Make webhook), `create_lead` in the CRM.
4. **Transfer**: warm transfer to the front desk number for emergencies or upset callers.
5. **Voice**: pick or clone a voice; set speed and interruption sensitivity.
6. **Phone number**: buy or import a number (Twilio/Plivo/Exotel) and attach the agent.
7. **Test**: 30 scripted calls (happy path, reschedule, noisy line, Hinglish, interruption, wrong date, emergency) before going live.
8. **Post-call**: send a WhatsApp/SMS confirmation and log a summary and outcome to the CRM.

Voice cloning

Type	Input	Use
Instant clone	About a minute of clean audio	Prototypes, personal use
Professional clone	30 minutes+ of studio-quality audio	Brand voice, courses, audiobooks
Voice design	A text description of the voice	When you do not want to clone a real person

WATCH OUT **Consent, disclosure and the law**

Clone only your own voice or a voice whose owner has given written permission. Disclose to callers that they are talking to an AI agent, follow telecom rules on automated and promotional calls (in India, TRAI's commercial-communication regulations and DND preferences), protect recordings under the DPDP Act, and never use cloned voices to impersonate or deceive. Reputable platforms require voice-verification for professional clones.

Measuring a voice agent

Metric	Target idea
Containment / resolution	Share of calls completed without a human
Booking or conversion rate	Appointments, payments or leads per call
Time to first audio, interruptions handled	Natural-feeling conversation
Transfer accuracy	Hands over the right calls
CSAT and complaint rate	Caller satisfaction
Cost per call vs human cost	Business case

Review call recordings and transcripts weekly, add failures to the test script, and update the prompt, knowledge and tools.

Summary

KEY TAKEAWAYS

- Voice agents chain telephony, speech-to-text, an LLM with tools and text-to-speech; latency and turn-taking make or break them.
- ElevenLabs, Vapi, Retell, Bland and Synthflow cover most needs; Twilio, Plivo and Exotel provide numbers.
- Build with a clear prompt, knowledge base, tools, transfer rules and a 30-call test script.
- Clone voices only with consent, disclose AI use, and follow telecom and data-protection rules.

Quick quiz

6 questions · tap an answer to check it

1. Why is latency critical for voice agents?

A Callers expect a reply within about a second; delays feel unnatural

B It affects storage cost only

C It changes the voice gender

D It does not matter

Answer: A. Stream every step and keep replies short.

2. What is barge-in?

A The caller interrupting and the agent stopping to listen

B A billing feature

C A voice clone

D A phone number type

Answer: A. Good turn-taking handles interruptions.

3. Which provides phone numbers and call routing?

A Telephony providers such as Twilio, Plivo or Exotel

B A vector database

C A spreadsheet

D An image model

Answer: A. Most voice platforms integrate them.

4. What must you have before cloning someone's voice?

- **A** Their explicit written consent
- **B** A paid plan only
- **C** A long prompt
- **D** Nothing

Answer: A. Also disclose AI use to listeners.

5. Which metric shows how many calls finish without a human?

- **A** Containment or resolution rate
- **B** Word count
- **C** Model size
- **D** Number of voices

Answer: A. Pair it with CSAT.

6. How do you connect a voice agent to your calendar if no native integration exists?

- **A** Call an n8n, Make or Zapier webhook as a tool
- **B** Read the calendar aloud
- **C** Email the caller a PDF
- **D** It is impossible

Answer: A. Webhooks bridge any app.

EXERCISE **Build it**

Create a free account on one voice platform and build the clinic receptionist (or a course-enquiry agent for SkillNavora). Connect a booking tool via native integration or an n8n webhook, write 15 test calls, and record the containment rate and average time to first audio.

Show solution (worked answer)

A good submission includes the prompt with rules and escalation, the knowledge file, the tool definitions, the transfer number, the test table (scenario, expected, actual) and two improvements made after testing.

INTERVIEW CORNER **Questions you may be asked**

Q: How would you launch a voice agent for customer support? Pick a narrow, high-volume use case; design the conversation, knowledge and tools; set transfer rules; test with scripted and real calls; launch to a small share of calls; monitor containment, CSAT and errors; iterate weekly.

Q: Speech-to-speech model or a pipeline? Speech-to-speech gives the lowest latency and natural prosody; a pipeline gives more control over each component, language choices, and tool use. Many teams start with a platform that manages the pipeline.

Q: What are the risks of voice cloning? Impersonation and fraud, consent and IP issues, regulatory problems with undisclosed AI calls, and reputational damage; mitigate with consent, verification, disclosure and watermarking.

Q: How do you support Indian languages? Choose STT and TTS that handle Hindi, Hinglish and regional languages, test accents, allow code-switching, and keep prompts and knowledge in the caller's language.

Q: What is the business case for a voice agent? Cost per resolved call vs human agents, 24×7 availability, faster answer times, conversion uplift, minus platform, telephony and maintenance costs, with quality guardrails.

Learn more

- ElevenLabs Agents documentation
- Vapi documentation
- Retell AI documentation

- Twilio voice quickstarts

Practise more online

Free, trusted places to test yourself on this topic. Links open the original site in a new tab.